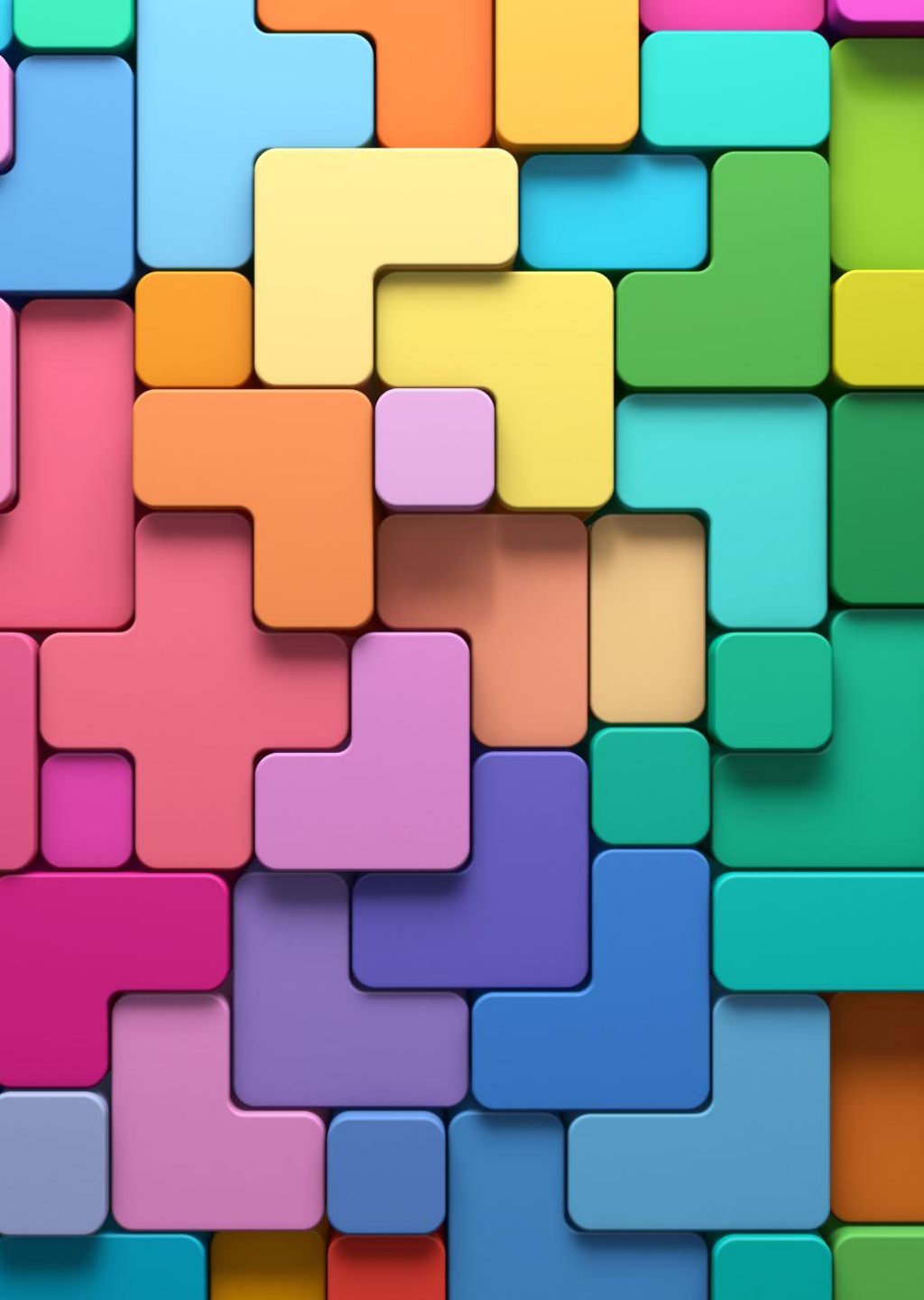


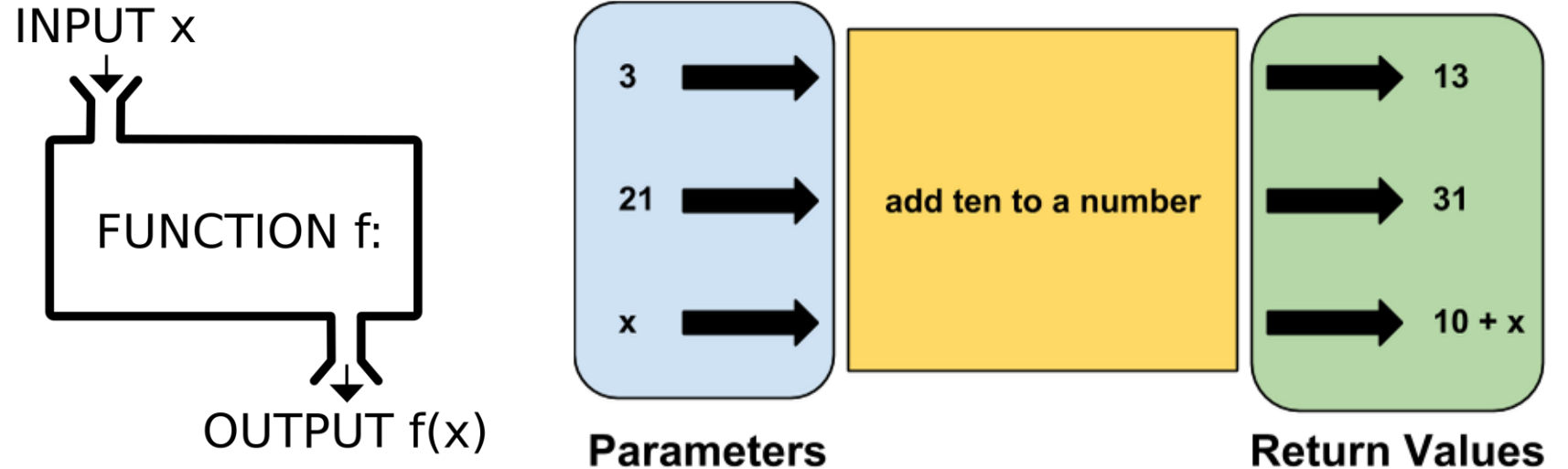


FUNCTIONS IN PYTHON

CSCI 1101 Introduction to Computer Science

WHAT IS A FUNCTION?

- A function is simply a “chunk” of code that you can use over and over again, rather than writing it out multiple times. A function's code performs a procedure or task and may return a value when completed.



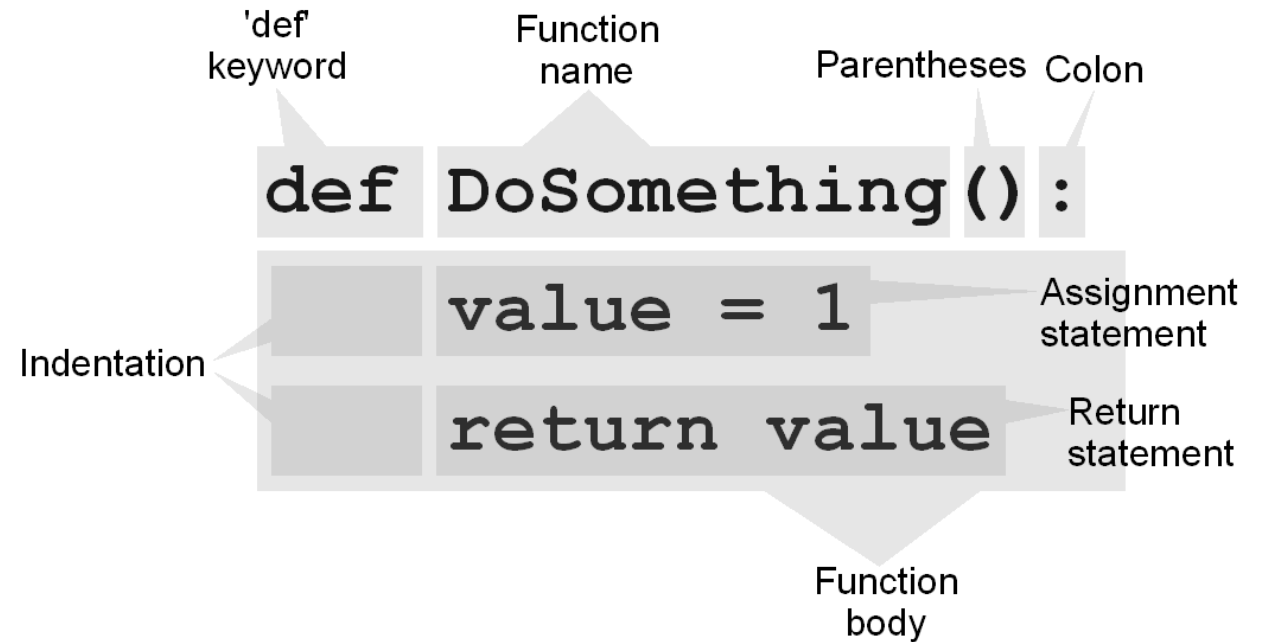
FUNCTIONS IN PYTHON

- Python provides us with many built in functions that would otherwise require extensive coding on our part.
- We have already used some functions in class such as `print()`, `input()`, `int()`, and `append()`.

The diagram shows the code `print("hello world")` with several red dashed boxes and labels explaining its parts:

- A box around the word `print` is labeled "Function Name".
- A box around the opening parenthesis `(` is labeled "Parentheses contain parameters".
- A box around the string `"hello world"` is labeled "Parameters separated by commas".
- A box around the closing parenthesis `)` is also labeled "Parameters separated by commas".

DEFINING A FUNCTION



- We define a function by using the **def** keyword and assigning a function name.
 - We can pass values to a function through its parenthesis `()`.
 - Any variables declared inside the function exist only inside the function body.
 - We can pass values outside the function by using a **return** statement.
-

INVOKING A FUNCTION

- Once a function has been defined, we can invoke it by calling its name. We must include the parenthesis when invoking a function.
- **Example**

`myVar = DoSomething()`

`print(myVar) -> 1`

`print( DoSomething() ) -> 1`
